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RESEARCH INTERESTS

1. Post-training Alignment for Trustworthy LLMs

Safety Alignment, Preference Learning

2. Robust and Reliable LLM Inference

Hallucination Mitigation, Jailbreak Detection, OOD Detection, Multimodal Robustness, Model Selection & Routing

3. Trustworthy LLM Agents and Agentic Systems

Agent Safety, Runtime Reliability, Multi-Agent Orchestration

EDUCATION

University of Southern California, Los Angeles, CA

Aug 2024 – Present

Ph.D. in Computer Science

· Affiliation: **FORTIS** Lab; Advisor: Prof. Yue Zhao

University of Southern California, Los Angeles, CA

Jan 2022 – Dec 2023

M.S. in Machine Learning and Data Science

Nankai University, Tianjin, China

Sep 2017 – Jul 2021

B.E. in Software Engineering

WORK EXPERIENCE

LinkedIn, AI/ML Engineer Intern, Generative AI

May 2026 – Aug 2026

User Simulation Agent, LLM Reasoning, Post-training

Sunnyvale, CA

- Developing the reasoning component of an LLM-based user-simulation agent to improve user-behavior prediction, via post-training with GRPO and OPSD.

SELECTED PUBLICATIONS

* denotes joint first authors. See [Google Scholar](#) for the full publication list.

· **Cat-DPO: Category-Adaptive Safety Alignment**

Tiankai Yang*, Yi Nian*, Xinyuan Li, Ruiyao Xu, Henry Peng Zou, Kaize Ding, Xiyang Hu, Yan Liu, Yue Zhao.
NeurIPS, 2026

· **HyperSkill: Self-Evolving LLM Agents via Hypergraph-Structured Skill Memory**

Ruiyao Xu, Tiankai Yang, Wei-Chieh Huang.
NeurIPS TTCL Workshop, 2026

- **SynCo: Data Synthesis Co-Training for Self-Evolving LLMs via Multi-Agent Reinforcement Learning**
Wei Yang, Li Li, Yuehan Qin, Yawei Wang, Mingxi Wang, Shixuan Li, Tiankai Yang, Jiate Li, Jesse Thomason, Xuezhe Ma, Yue Zhao.
NeurIPS TTCL Workshop, 2026
- **CoAct: Co-Active LLM Preference Learning with Human-AI Synergy**
Ruiyao Xu, Mihir Parmar, Tiankai Yang, Zhengyu Hu, Yue Zhao, Kaize Ding.
ACL, 🏆 Oral, 2026 🔄
- **DOG-DPO: Dynamic Optimization in Geometry for Safety Alignment**
Yi Nian, Tiankai Yang, Yudi Zhang, Qi Pan, Zelong Xu, Shenzhe Zhu, Qingqing Luan, Yue Huang, Xiangliang Zhang, Yue Zhao.
EMNLP Findings, 2026
- **FlexRouter: Learning Complementary Model Sets for Flexible LLM Routing**
Wang Wei, Harry Yang, Tiankai Yang, Samyadeep Basu, Hongjie Chen, Andy Zhao, Franck Dernoncourt, Ryan A. Rossi, Hoda Eldardiry.
COLM, 2026
- **AD-AGENT: A Multi-agent Framework for End-to-end Anomaly Detection**
Tiankai Yang*, Junjun Liu*, Michael Siu*, Jiahang Wang, Zhuangzhuang Qian, Chanjuan Song, Cheng Cheng, Xiyang Hu, Yue Zhao.
IJCNLP-AACL Findings, 2025 🔄
- **AD-LLM: Benchmarking Large Language Models for Anomaly Detection**
Tiankai Yang*, Yi Nian*, Shawn Li, Ruiyao Xu, Yuangang Li, Jiaqi Li, Zhuo Xiao, Xiyang Hu, Ryan A. Rossi, Kaize Ding, Xia Hu, Yue Zhao.
ACL Findings, 2025 🔄
- **DPU: Dynamic Prototype Updating for Multimodal Out-of-Distribution Detection**
Shawn Li, Huixian Gong, Hao Dong, Tiankai Yang, Zhengzhong Tu, Yue Zhao.
CVPR, 🏆 Highlight, 2025 🔄
- **PyOD 2: A Python Library for Outlier Detection with LLM-powered Model Selection**
Sihan Chen*, Zhuangzhuang Qian*, Michael Siu*, Tiankai Yang, Xingcan Hu, Jiaqi Li, Shawn Li, Yuehan Qin, Zhuo Xiao, Wanghao Ye, Yichi Zhang, Yushun Dong, Yue Zhao.
The Web Conference (Demo Track), 2025 🔄

SELECTED PREPRINTS & UNDER SUBMISSION

* denotes joint first authors. See *Google Scholar* for the full list.

- **Model Routing for Large Language Models: A Unified Taxonomy and Survey**
Wang Wei, Tiankai Yang, Xiangbo Gao, Chan-Wei Hu, Reuben Luera, Runzhou Liu, Nigel Fernandez, Li Li, Lunyiu Nie, Bo Ni, Junda Wu, Chien Van Nguyen, Deonna Owens, Hongjie Chen, Seunghyun Yoon, Puneet Mathur, Tong Yu, Sungchul Kim, Koustava Goswami, Zichao Wang, Joe Barrow, Samyadeep Basu, Xiang Chen, Namyong Park, Nesreen K. Ahmed, Nedim Lipka, Thien Huu Nguyen, Ting-Hao Kenneth Huang, Franck Dernoncourt, Trung Bui, Ryan A. Rossi, Yue Zhao, Zhengzhong Tu, Hongru Du, Tyler Derr, Yu Wang, Hoda Eldardiry.
Under submission. SSRN preprint 🔄
- **No Attacker Needed: Unintentional Cross-User Contamination in Shared-State LLM Agents**
Tiankai Yang, Jiate Li, Yi Nian, Shen Dong, Ruiyao Xu, Ryan A. Rossi, Kaize Ding, Yue Zhao.
Under submission. arXiv preprint arXiv:2604.01350

SERVICES

Program Committee for Conferences

- NeurIPS 2026; ACL ARR 2025 (*Outstanding Reviewer*), 2026; AAAI 2026, 2027; COLM 2026; ICLR 2026, 2027; ICML 2025, 2026.

Journal Reviewer

- IEEE Transactions on Computational Social Systems